Extra Bite #9: Yagis, Parabolic Dishes, and Electrically Short Antennas

Yagi mechanics, dish gain math, and what happens when an antenna is too short for its band.

Group E9D · 12 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

Yagi Antennas

Question	Answer	Q
How long is a Yagi's driven element?	About **1/2 wavelength**. The standard half-wave dipole length is the starting point.	E9D05
Why are parasitic elements made slightly longer or shorter than resonance?	For **control of phase shift**. A longer parasitic looks inductive and re-radiates with a phase lag (reflector behavior); shorter looks capacitive and leads (director behavior).	E9D12
Why does a typical 2-element Yagi have a reflector instead of a director?	**Higher gain** at normal element spacing. (E9D11)	E9D11
How to make circular polarization from two linear Yagis?	**Arrange two Yagis on the same axis and perpendicular to each other**, with the driven elements at the same point on the boom, and feed them **90° out of phase**. (E9D02)	E9D02

Parabolic Reflectors (Dishes)

Double the operating frequency and an ideal parabolic dish gains +6 dB. (E9D01)
Reasoning: dish gain depends on the dish's size in *wavelengths*. Doubling the frequency halves the wavelength, so the same physical dish is twice the diameter in wavelengths – four times the captured area – which is +6 dB.

Electrically Short Antennas

A whip or vertical that is significantly shorter than 1/4 wavelength at the operating frequency presents a low resistance plus a big capacitive reactance. To make it useful, the capacitive reactance has to be cancelled and the SWR brought into range.

Loading coils

Question	Answer	Q
Function of a loading coil	**To resonate the antenna by cancelling the capacitive reactance.** (E9D09)	E9D09
Most efficient location on a short whip	**Near the center of the vertical radiator.** (E9D03) Center loading lets the coil sit where the current is still high but not at the peak; gives the best compromise between efficiency and practicality.	E9D03
Why high reactance-to-resistance ratio (high Q) coils?	**To maximize efficiency.** (E9D04) Any resistance in the coil burns transmitter power as heat instead of radiating it.	E9D04
SWR bandwidth after adding loading coils?	**Decreases.** (E9D06) The loaded antenna is sharper.	E9D06
Top loading advantage	**Improved radiation efficiency.** (E9D07) Capacitive top hats raise the current distribution along the radiator without adding lossy coil resistance.	E9D07

Antenna Q and radiation resistance

Question	Answer	Q
As antenna Q increases	**SWR bandwidth decreases.** (E9D08) The narrow-banded antenna is just the high-Q tank circuit it is.	E9D08
Radiation resistance of a base-fed whip below its resonant frequency	**Decreases.** (E9D10) Shorter-than-resonant antennas put less of their feed power into actual radiation and more into the reactive component, so the radiation resistance drops.	E9D10

Mental Cheat Sheet

- Yagi driven element: about 1/2 wavelength.
- Parasitic elements longer/shorter than resonance = phase shift control.
- 2-element Yagi with reflector: higher gain.
- Circular polarization from two Yagis: same axis, perpendicular, same boom point, 90° out of phase.
- Parabolic dish: **+6 dB** when frequency doubles.
- Loading coil = cancel capacitive reactance to resonate.
- Best loading-coil location on a short whip: **center**.
- High coil Q (reactance/resistance ratio) = high efficiency.
- Adding loading coils ⇒ SWR bandwidth **decreases**.
- Top loading ⇒ improved radiation efficiency.
- Higher antenna Q ⇒ SWR bandwidth **decreases**.
- Base-fed whip below resonance ⇒ radiation resistance **decreases**.

Common Traps

- Doubling the frequency of an ideal parabolic dish adds **6 dB**, not 3 or 4.
- Loading coils on a short whip work best **near the center**, not at the base or top.
- The point of high-Q loading coils is **efficiency**, not bandwidth or harmonic suppression.
- Adding loading coils makes SWR bandwidth **narrower**, not wider.
- Top loading improves **efficiency**, not gain or front-to-back ratio.
- Parasitic-element length shift is for **phase shift** control, not mechanical balance or wind torque.

Takeaway: Yagis use parasitic elements whose length shift controls phase to create reflector and director behavior; a 2-element Yagi gets the most gain from a reflector. Parabolic dishes gain 6 dB per frequency doubling. Electrically short antennas need loading coils (best at the center, ideally high-Q) to cancel capacitive reactance, but the price is narrower SWR bandwidth; the lower radiation resistance is an intrinsic property of the short antenna, not a consequence of the loading coil.

Bite 9 Question Bank (E9D)

E9D01. How much does the gain of an ideal parabolic reflector antenna increase when the operating frequency is doubled?
A. 2 dB.
B. 3 dB.
C. 4 dB.
D. 6 dB.

E9D02. How can two linearly polarized Yagi antennas be used to produce circular polarization?
A. Stack two Yagis to form an array with the respective elements in parallel planes fed 90 degrees out of phase.
B. Stack two Yagis to form an array with the respective elements in parallel planes fed in phase.
C. Arrange two Yagis on the same axis and perpendicular to each other with the driven elements at the same point on the boom and fed 90 degrees out of phase.
D. Arrange two Yagis collinear to each other with the driven elements fed 180 degrees out of phase.

E9D03. What is the most efficient location for a loading coil on an electrically short whip?
A. Near the center of the vertical radiator.
B. As low as possible on the vertical radiator.
C. At a voltage maximum.
D. At a voltage null.

E9D04. Why should antenna loading coils have a high ratio of reactance to resistance?
A. To swamp out harmonics.
B. To lower the radiation angle.
C. To maximize efficiency.
D. To minimize the Q.

E9D05. Approximately how long is a Yagi's driven element?
A. 234 divided by frequency in MHz.
B. 1005 divided by frequency in MHz.
C. 1/4 wavelength.
D. 1/2 wavelength.

E9D06. What happens to SWR bandwidth when one or more loading coils are used to resonate an electrically short antenna?
A. It is increased.
B. It is decreased.
C. It is unchanged if the loading coil is located at the feed point.
D. It is unchanged if the loading coil is located at a voltage maximum point.

E9D07. What is an advantage of top loading an electrically short HF vertical antenna?
A. Lower Q.
B. Greater structural strength.
C. Higher losses.
D. Improved radiation efficiency.

E9D08. What happens as the Q of an antenna increases?
A. SWR bandwidth increases.
B. SWR bandwidth decreases.
C. Gain is reduced.
D. More common-mode current is present on the feed line.

E9D09. What is the function of a loading coil in an electrically short antenna?
A. To increase the SWR bandwidth by increasing net reactance.
B. To lower the losses.
C. To lower the Q.
D. To resonate the antenna by cancelling the capacitive reactance.

E9D10. How does radiation resistance of a base-fed whip antenna change below its resonant frequency?
A. Radiation resistance increases.
B. Radiation resistance decreases.
C. Radiation resistance becomes imaginary.
D. Radiation resistance does not depend on frequency.

E9D11. Why do most two-element Yagis with normal spacing have a reflector instead of a director?
A. Lower SWR.
B. Higher receiving directivity factor.
C. Greater front-to-side.
D. Higher gain.

E9D12. What is the purpose of making a Yagi's parasitic elements either longer or shorter than resonance?
A. Wind torque cancellation.
B. Mechanical balance.
C. Control of phase shift.
D. Minimize losses.

Answer Key

E9D01: D	E9D02: C	E9D03: A	E9D04: C
E9D05: D	E9D06: B	E9D07: D	E9D08: B
E9D09: D	E9D10: B	E9D11: D	E9D12: C